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Transforming cold chain logistics: a reversible vehicle routing approach for sustainable and efficient delivery of perishable goods

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ABSTRACT

As global demand for perishable goods rises, logistics providers face challenges in managing temperature-sensitive products. Traditional vehicle routing models often neglect cold chain requirements, including temperature control and strict delivery schedules. Additionally, sustainability concerns drive the need for eco-friendly logistics practices, as passive packaging methods contribute to waste. This study introduces a reversible cold chain vehicle routing model that integrates the Internet of Things and a multi-objective optimisation framework for real-time monitoring. Incorporating cold chain time windows, recycling constraints, and social sustainability, the model demonstrates how technology can enhance cold chain logistics and support circular economy goals.

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KEYWORDS

Cold chain logistics; vehicle routing; social sustainability; recyclable passive packaging; temperature-sensitive products

1. Introduction

In recent years, research related to cold chains has received continuous interest, focusing on the handling process of temperature-sensitive and perishable products, which includes critical stages such as precooling, storage, packing and delivery. Unlike standard supply chains, the cold chain environment is specifically designed to protect and maintain the quality of temperature-sensitive items during transportation, ensuring that they preserve their shelf life and reach consumers globally across various locations. Hence, it is imperative to implement effective operational processes and utilise suitable materials to handle these perishable products efficiently. Neglecting these factors can lead to contamination or loss during transit, ultimately resulting in substantial economic waste (Ni et al. 2022).

Along a cold chain, packaging is considered crucial for essential functions, such as protection, containment, temperature control and unitisation (Pålsson 2018). Although the primary goal is often to minimise losses, packaging practices, especially the selection of appropriate packaging materials, play a crucial role in waste reduction (Conte et al. 2015; J. Wang et al. 2019). Currently, cold chain logistics (CCL) utilises two main types of packaging. The first is active packaging, which involves systems that rely on an external

energy source, such as refrigeration, to maintain the desired temperature range for products during distribution and storage (Ahmed et al. 2017). Although this method is generally more expensive, it offers more reliable temperature control and is better suited for long-distance delivery. The second is passive packaging, which does not depend on an external energy source to maintain the required temperature range. Instead, it uses insulation materials and innovative designs to sustain the necessary temperatures over a defined period without active cooling systems. This approach reduces the reliance on continuous refrigeration, thereby significantly lowering energy consumption (Chandran et al. 2022). Although typically more cost-effective than active packaging methods, passive packaging methods have limitations such as less precise temperature control and reduced suitability for long-distance shipment. However, its lower cost and greater flexibility make it a popular choice in cold chain transportation, particularly among small- and medium-sized logistics providers. Nevertheless, the use of passive materials raises environmental concerns because they can contribute to solid waste if not effectively recycled after a single shipment. The concept of reversible packaging has gained attention in addressing this problem. The design allows for multiple uses and promotes its repurposing or reconditioning for subsequent shipments. This approach not only extends the lifecycle of packaging materials in CCL but also diminishes the demand for new resources, thereby reducing the overall environmental footprint (Jiang et al. 2023).

To address both logistics and sustainability challenges within cold chains, previous studies have explored various vehicle routing problems (VRPs), including variants such as the capacitated VRP (CVRP), VRP with time windows (VRPTW) and pickup and delivery VRP (VRPPD) (Pillac et al. 2013). The emergence of the green VRP (GVRP) has introduced environmental considerations into route planning, across aspects such as selective pickups with pricing, waste collection, end-of-life goods collection and simultaneous distribution and collection (Lin et al. 2014). Ganji et al. (2020) studied a multi-objective scheduling problem that considers various factors, including transportation costs, fixed costs, fuel costs, carbon emissions, delays and customer dissatisfaction. G. Liu et al. (2020) explored a joint distribution-GVRP (JD-GVRP) that integrates fixed, transportation, damage, refrigeration, penalty and carbon costs into its framework. Furthermore, Li, Lim, and Tseng (2019) proposed a green vehicle routing model based on a modified particle swarm optimisation tailored specifically for CCL, addressing vehicle operating, quality loss, product freshness, penalty, energy and greenhouse gas emission costs. Despite significant advancements in CCL, several critical areas remain unexplored. Although various VRP variants exist, there is a notable gap in research that specifically focuses on green vehicle routing challenges within cold chain contexts. He et al. (2018) emphasised that data mining techniques could enhance the collection and analysis of logistics data, which is vital for developing more accurate and responsive routing models. They also emphasised the disparity between theoretical models and their practical applications in real-world settings. Moghdani et al. (2021) highlighted numerous opportunities for future research on GVRPs in areas such as problem classification, consideration of uncertainties, solution methodologies and balancing single and multiple objectives.

To address these research gaps, this study proposed a reversible cold chain vehicle routing problem (RCCVRP) model. The novelty of this research is twofold: (i) a multi-objective optimisation framework that simulates the complexities associated with reverse logistics in cold chain operations was developed, and (ii) the objectives were evaluated

based on a range of parameters spanning the economic, cold chain and sustainability dimensions. As a result, this study aims to empower practitioners to effectively develop, modify and analyse scheduling schemes, thereby enhancing the efficiency of CCL operations while concurrently improving customer satisfaction and fostering environmental responsibility.

The remainder of this paper is structured as follows. [Section 2](#) presents a literature review on the cold chain, sustainability and VRPs. [Section 3](#) presents the methodology of the study. [Section 4](#) and [5](#) present a case study and the results and discussion, respectively. Finally, [section 6](#) concludes the paper.

2. Literature review

2.1. Cold chain logistics

The cold chain field was developed to handle environmentally sensitive products (EPS) and maintain product quality and safety (Tsang et al. 2017). It is a system for handling perishable items and sensitive goods that must be in a specific temperature range during delivery, such as fruits, vegetables and vaccines, to increase their shelf life (Patidar, Shukla, and Sukhwani 2022). A cold chain can be divided into four parts: precooling, refrigeration, transportation and marketing (Fu et al. 2024). Each component is correlated and interdependent, and problems or issues within the chain can cause losses and waste. For example, the quality of temperature-sensitive products affects final product prices (Nakandala, Lau, and Zhang 2021). The effectiveness of CCL depends on three main factors: properties, infrastructure maintenance and modernity. The important elements of infrastructure and modernity include storage, transportation and packaging materials (Han et al. 2021). Currently, cold chains are commonly used to extend the shelf life of perishable foods, enabling customers to obtain high-quality and safe products (Loisel et al. 2021). Traditional challenges in cold chain systems include cold chain capacity, equipment and temperature control. The inadequacy of temperature management and monitoring can lead to unusable products and hence wastage and increased carbon emissions, which can have grave consequences in cases such as vaccinations in the developing world (Ashok, Brison, and LeTallec 2017; Tsai and Lin 2019). Therefore, several enterprises have implemented advanced technology such as blockchain and the Internet of Things (IoT) to monitor delivery environments (Srivastava, Zhang, and Eachempati 2023). IoT applications can be divided into three different layers: the perception, network and application layers (Swamy, Jadhav, and Kulkarni 2017). In addition, CCL is highly energy-consuming and hence highly carbon-emitting. Therefore, considerable research has been conducted to balance economic, cold chain and environmental concerns (G. Liu et al. 2020; Qin, Tao, and Li 2019; S. Wang, Tao, and Shi 2018; S. Wang et al. 2017). Currently, the focus of CCL is not only on maintaining product quality and safety through effective temperature control but also on the growing attention to environmental responsibility and sustainability (Mo et al. 2024). While effective CCL is important for monitoring and maintaining temperature-sensitive product quality and safety, it increases the environmental impacts of these necessary operations

during transportation, such as in the higher consumption of carbon emissions and the ecological footprint of the CCL process. Therefore, a deeper exploration of sustainability practices in this field is necessary.

2.2. Sustainability in cold chain logistics

Sustainability is a type of stability characterised by a quantity that remains invariant amidst balancing between a population; in other words, 'a dynamic equilibrium in the process of interaction between a population and the carrying capacity of its environment such that the population develops to express its full potential without producing irreversible adverse effects on the carrying capacity of the environment upon which it depends' (Ben-Eli, 2018). Research on 'sustainability' has gained attention over the past two decades and can be broadly divided into three aspects: social, economic and environmental, which overlap and interact with one another (Purvis, Mao, and Robinson 2019). The triple bottom line (TBL) theory is an iconic conceptual framework for evaluating sustainability performance and the overall impact of an organisation's activities. It emphasises that businesses should focus on three key areas: the economic, environmental and social dimensions (Nogueira, Gomes, and Lopes 2023). The TBL encourages organisations to incorporate social and environmental considerations into their core business strategies. This leads to more informed decision making that balances economic performance with societal and ecological impacts. These three dimensions can be divided into factors such as revenue, profitability and cost from an economic perspective; energy consumption, disposal and greenhouse gas discharge from an environmental perspective; and employee welfare, customer satisfaction and ethical business conduct from a social perspective (Sahoo and Upadhyay 2024). A primary concern promoting sustainability is climate change, which is exacerbated by the use of fossil fuels. In particular, transportation is a significant consumer of energy and hence a source of greenhouse gas emissions (F. Liu et al. 2018). Carbon dioxide is the primary greenhouse gas produced by vehicles, accounting for 80% of greenhouse gas emissions. Owing to its role in climate change, tackling its emission is vital for economic and environmental sustainability (Ashena and Shahpari 2022). Storage and transportation are the two major factors that drive the environmental sustainability of CCL (Kumar, Tyagi, and Sachdeva 2023). For example, energy usage, water usage, paper usage and material wastage are sources of carbon emissions in CCL (Tsang, Ma, et al. 2024). Time and temperature management is a point of penetration for CCL to achieve sustainability by balancing cold chain performance and costs (Chen et al. 2022). To achieve sustainability in CCL, effective packing solutions and strategies are required to reduce energy consumption and greenhouse gas emissions during storage and transportation, while maintaining the efficiency and resilience of the supply chain. Social sustainability considers factors related to workforce welfare, such as reduction in worker variations, whereas economic sustainability considers factors related to reduced waste and improved logistics, such as cost increases (Chaabane et al. 2022). To achieve sustainability in CCL, reducing carbon emissions and optimising packaging and routing strategies are essential. Furthermore, economic optimisation and social reasonability should be considered to ensure environmental, social and economic balance.

2.3. Packaging practices in cold chains

Under cold chain management, packaging plays an important role in product quality and shelf life. Cold chain packaging has been applied in several fields, including the food, meat and pharmaceutical industries, to maintain shelf lives and provide higher-quality products (Nastasijević, Lakićević, and Petrović 2017; Yildirim et al. 2018). Typically, packaging can be classified into three types based on its layers and functions: primary packaging, which directly envelops products, secondary packaging, which is placed outside the primary packaging and tertiary packaging, which is used when delivering the products (Mahmoudi and Parviziomran 2020). In CCL, refrigeration packaging can be arranged for secondary and tertiary packaging to handle and provide temperature-sensitive goods following appropriate requirements. Refrigeration packaging can be divided into two types: i) active packaging, which requires a thermostatic-controlled energy source system to maintain the temperature of the product, and ii) passive packaging, which uses conventional packing combined with dry ice and phase change materials (PCMs) to mitigate the impact of the external temperature on products (Sykes 2018). Passive packaging is commonly used in cold chain management owing to its low cost, low energy consumption and high transportation volume (Sepe, Armentani, and Pozzi 2015). Another area that has gained attention is waste in packaging. A recent research trend involves the creation of renewable packaging solutions, such as recyclable packaging materials. This has resulted in a search for environmentally friendly solutions, which require further research (Han et al. 2018). Some studies have been conducted to include biologically inspired smart packaging and biodegradable materials such as bioplastic polymers (Mlalila et al. 2016). Another branch of research focuses on consumer behaviour regarding packaging and addresses the issue of overpackaging (Monnot et al. 2019). Reverse logistics processes have a significant impact on sustainability by reducing the recycling of packaging materials. With an effective reverse logistics policy, reusable packaging materials can be collected and returned immediately or the next day for a subsequent delivery (Lai et al. 2022). To optimise the benefits of reverse logistics in cold chain operations, it is imperative to optimise strategic delivery routing to minimise costs and effectively collect reusable packaging materials. Tsang, Fan, et al. (2024) noted that the effectiveness of optimising packaging methods depends on delivery routing. Therefore, the integration of the cold chain VRP can effectively balance cost, quality, sustainability and reversible packaging use.

2.4. Vehicle routing problems (VRPs) in cold chain logistics

The ‘truck dispatching problem’, or the more general ‘VRP’ has been studied since the 1960s to address the economic and environmental significance of transportation (Braekers, Ramaekers, and Van Nieuwenhuyse 2016). The VRP is defined as an optimisation problem that is applied to determine the best delivery and collection routing for delivering goods from one or multiple locations to customers in different places on a geographical scatter (Rahmani, Ramdane Cherif-Khettaf, and Oulamara 2016). Subsequently, numerous variants of the VRP have been studied. Some of the most popular and widely used variants are the VRPB, VRPTW and VRPPD (Konstantakopoulos, Gayialis, and Kechagias 2022). Owing to its inherent

characteristics, CCL has gained significant attention from researchers. Unlike those of traditional logistics, the goods of CCL are temperature- and time-sensitive (Theophilus et al. 2021). Therefore, it requires the delivery of goods to the end customer in high-quality condition under the required storage temperature and within an optimal time window. An early application of the VRP to a cold chain considered a cold chain time window (Hsu, Hung, and Li 2007). Other variants of the VRP have been applied to CCL. For example, Gong and Liu (2019) implemented the VRPB approach by considering the cost, carbon emissions and time window in cold chain delivery. Zhang et al. (2020) applied the VRPTW to CCL by considering the time window constraints required by customers. Pilati and Tronconi (2024) applied the VRPPD to consider optimal sustainable routing. GVRPs are a new variant of the VRP that considers environmental sustainability issues (Fernández Gil et al. 2022). These problems are aimed at identifying effective routes with fewer environmental concerns. While previous research has explored the GVRP in cold chains with environmental metrics, studies focusing on integrating reverse logistics and addressing multi-objective problems in cold chain contexts are still limited. Incorporating sustainability and packaging considerations into the VRP model can help to arrive at cost-efficient, convenient, socially responsible and environmentally sustainable routes. This integration requires a multi-objective model to address the trade-off.

2.5. Summary

To summarise, CCL and sustainability are two major areas of research in logistics operations, and the VRP, which is a popular research area, is used to address both simultaneously. As stated in the Introduction, there is limited research on multi-objective optimisation, data collection and practical applications that integrate environmental concerns when handling temperature-sensitive products in routing. The methodology is divided into two parts: the first involves the RCCVRP model, a multi-objective variant of the VRP with considerations of i) labour cost, ii) fuel cost, iii) transit carbon emission, iv) idle carbon emission and v) fixed cost, which performs cold chain routing using recyclable passive packaging and the resultant reverse logistics for recycling. The second part provides a practical methodology that determines the objectives of the optimisation function using the collected data.

3. Methodology

To provide an effective CCL service, practitioners must balance performance cost with satisfying the constraints of customer and quality demand. Furthermore, in this age of social responsibility, practitioners must also consider sustainability. This study investigated the use of recyclable passive packaging and its logistics operation by first modelling and then solving multi-objective optimisation with constraints as a variant of the RCCVRP.

3.1. Operational model of closed-loop cold chain logistics

In practice, the primary functions of a typical transportation management system (TMS) are to (i) optimise routing and scheduling of transportation content, (ii) plan how to load shipments efficiently and (iii) track shipments to enhance efficiency, reduce costs and improve service levels in transportation management. This study proposed the RCCVRP model for CCL and enhanced the existing TMS to (i) optimise the routing problem by considering multiple objectives to simulate the complexities of reverse logistics in cold chain content with environmental concerns, (ii) optimise load planning to ensure that the right products are loaded by considering the specific temperature needs of different products and (iii) enable real-time tracking of the temperature of sensitive shipments using IoT technology. Compared to general product shipments, CCL requires more product protection and monitoring of environmental conditions for delivery. Therefore, additional cold chain packaging and monitoring devices are involved in these operations. However, most passive packaging and sensor tags are designed for one-time use to eliminate the reverse logistics delivery and recycling costs. Owing to innovations in packaging materials and increased consideration of sustainable development, new packaging materials and monitoring devices have been designed for reusability. Thus, there is a gradual shift to considering a more eco-friendly and reusable solution to reduce the environmental impact and enhance economic efficiency. This study considered a closed-loop CCL model, as shown in Figure 1. CCL service providers (CCLSPs) have a set of transportation teams for delivery of shipments and collection of used packaging materials. To protect products based on customer requirements, they are delivered using appropriate packaging and monitoring devices, such as polyfoam boxes, eutectic plates, data tags and sensors. Moreover, transportation teams collect cold chain packaging materials upon delivery and when passing through shops or depots. These used packaging materials undergo disinfection and functional checking before being reused in the subsequent distribution.

Furthermore, the IoT is adopted in route planning to facilitate product monitoring and optimal delivery planning in CCL. The detailed IoT application layers are shown in Figure 2, which consist of the perception, network and application layers. The IoT

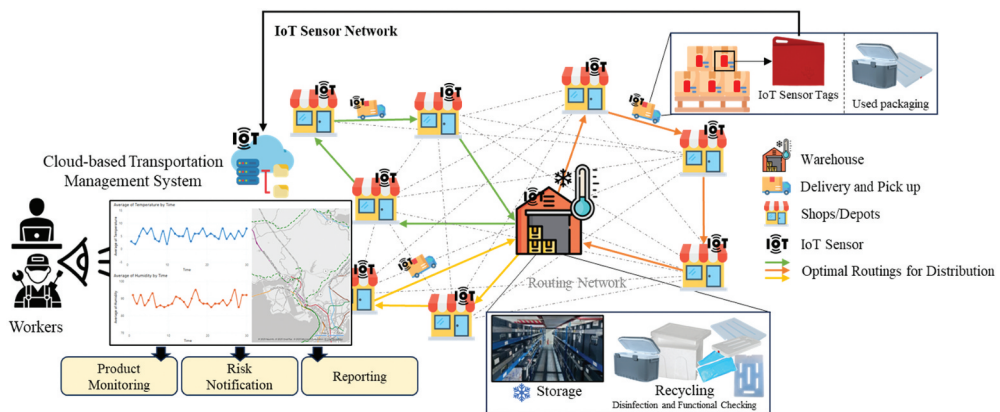


Figure 1. Operations of closed-loop cold chain logistics network.

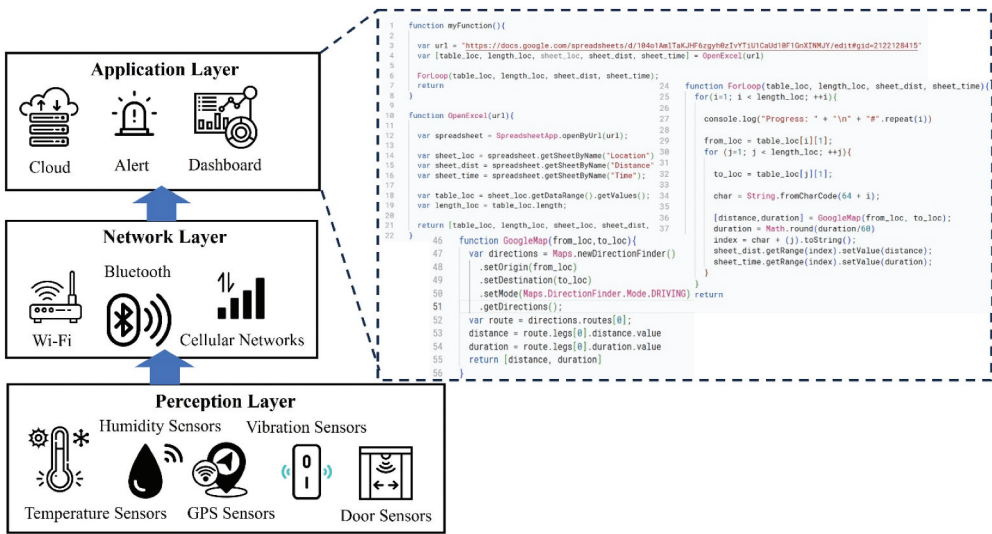


Figure 2. Key components of IoT layers in cold chain management.

improves the visibility and traceability of environmental conditions, such as temperature and humidity, during transportation. Sensing technology uses sensor tags to collect environmental data, provide product information and identify passive packaging box information. During transportation, the sensor operates in a wireless local area network (WLAN) to collect the temperature, humidity and product information. Data can be transmitted to a Platform as a Service (PaaS) for management through a cloud configuration, including a cloud-based TMS. Subsequently, a cloud database can manage and structure the transmitted data. Finally, these structured data can be used by workers in business intelligence applications for real-time data visualisation, such as product consideration monitoring, risk notification and reporting. Such enhancement provides workers and managers with real-time tracking in transportation management. This allows them to understand the distribution process and routings and react to risk incidents effectively, strengthening their exception management. Optimal CCL operations aim to achieve sustainable development, with a focus on enhancing environmental sustainability, economic efficiency and operational effectiveness. In addition, GPS sensors provide real-time vehicle locations for route planning. If there is a sudden traffic jam or unfavourable road conditions, the distance and time matrix can be updated instantly from the Google Maps API and the routing plan can be revised simultaneously to advance transportation planning and scheduling. Therefore, the RCCVRP model integrates real-time temperature monitoring, dynamic routing suggestions and real-time data visualisation to improve the overall operation and enhance the TMS.

3.2. Model description

3.2.1. Assumptions and notations

In cold chain operations with passive packaging, end customers can make one of two requests: (i) order requests, in which cold chain products are delivered from the depot,

and (ii) recycling requests, where the passive packages used for previous deliveries are returned to the logistics service provider (LSP). The following six assumptions were adopted in this model:

- (i) Each vehicle starts and ends each trip at the depot.
- (ii) All the goods to be delivered are ready for departure from the depot before the trip begins.
- (iii) Each labourer is paid at a fixed hourly rate, and thus the labour cost is bilinear in time and labour pool size.
- (iv) Using a vehicle incurs a fixed cost, which includes costs of truck wear and tear, maintenance and carbon emissions for starting the engine.
- (v) Carbon emissions in transit are modelled as linear with travel distance.
- (vi) Carbon emissions when idle are modelled as linear with slack time.

Based on these assumptions, the routing process is represented as a set of action nodes that must be completed, and a set of vehicles is employed to complete them. The notations in Table 1 represent the nature of each order (for example, a delivery request or a recycling request) and the decision variables.

Table 1. Notations of the parameters and decision variables.

Notation	Description
Parameters	
n	The number of nodes
$N = \{1, 2, \dots, n\}$	Index for nodes
h	The number of vehicles
$V = \{1, 2, \dots, h\}$	Set of vehicles
u_i^G	Volume of goods delivered at node i
u_i^P	Volume of packaging materials picked up at node i
w_i^G	Weight of goods delivered at node i
w_i^P	Weight of packaging materials picked up at node i
ET_i^C	Earliest customer available time for node i
LT_i^C	Latest customer available time for node i
ET_i^{CC}	Earliest cold chain time window for node i
LT_i^{CC}	Latest cold chain time window for node i
d_{ij}	Travel distance from node i to j
t_{ij}	Expected travel time from node i to j
V_v	Volume capacity of the cargo of vehicle v
W_v	Weight capacity of the cargo of vehicle v
L_v^{max}	Maximum number of workers in vehicle v
V_t^{max}	Maximum volume goods or packaging materials that can be served per unit time
V_i^{max}	Maximum volume of goods or packaging materials that can be handled per worker
DU_i^v	Cumulative delivery volume of vehicle v after serving node i
PU_i^v	Cumulative pickup volume of vehicle v after serving node i
DW_i^v	Cumulative delivery weight of vehicle v after serving node i
PW_i^v	Cumulative pickup weight of vehicle v after serving node i
TD_i^v	Cumulative travel distance of vehicle v after arriving at node i
TT_i^v	Cumulative travel time of vehicle v after arriving at node i
s_i^v	Slack time of vehicle time for vehicle v stopping at node i
L_v	Number of workers required by vehicle v
Decision Variables	
X_{ij}^v	Binary variable denoting routing sequence of vehicle v
Y_v	Binary variable denoting occupancy of vehicle v
u_i^v	Volume carried by vehicle v after serving node i
w_i^v	Weight carried by vehicle v after serving node i

3.2.2. Model objectives

The model in this study is an extended version of an earlier model presented in a conference paper (Lam, Tang, and Ho 2023). In this extended version, additional factors, including social sustainability, are considered. Equation (1) is an objective function considering multiple factors with the aim of minimising costs incurred from labour, fuel consumption, carbon emissions from travelling, and idle carbon emissions as well as the fixed cost incurred by each vehicle, while maximising social sustainability in terms of labour welfare. Generally, $\pi_1 - \pi_5$ are cost factors, whereas π_6 is a benefit factor. The labour factor π_1 states that the labour cost is proportionate to the number of labourers as well as the time incurred, as described in Eq. (2). The fuel consumption factor π_2 states that the fuel cost is proportional to the total travel distance, as described in Eq. (3). The factor of carbon emission from travelling π_3 states that the carbon emission is proportional to the total travel distance, as described in Eq. (4). The factor of idle carbon emission π_4 states that the idle carbon emission is proportional to the slack time, as described in Eq. (5). The factor of fixed cost π_5 states that the fixed cost incurred by each vehicle is proportional to the number of vehicles employed, as described in Eq. (6). Moreover, social sustainability in terms of labour welfare is calculated using Eq. (7), which is equal to the total number of labourers required divided by the number of assigned vehicles. This implies that increasing the labour assigned per vehicle can improve working conditions, yielding greater social benefits. By incorporating π_6 in the denominator of the overall objective function, the formula transforms the maximisation of social sustainability into a minimisation problem. λ_p is introduced to adjust the unit conversion rate of each cost factor. Owing to the difference in importance of the factors, two variables, α and β , are included in the objective function to adjust the weightings of the cost factors and social benefit factors, respectively. The values of the two variables are determined by the stakeholders' strategic priorities and focus.

$$Min \alpha \sum_{p=1}^5 \lambda_p \pi_p / \beta \pi_6 \quad (1)$$

$$\pi_1 = \sum_{v \in V} C_v^L L_v^V T_{n+1}^V \quad (2)$$

$$\pi_2 = \sum_{v \in V} C_v^F T D_{n+1}^V \quad (3)$$

$$\pi_3 = \sum_{v \in V} E_1^V T D_{n+1}^V \quad (4)$$

$$\pi_4 = \sum_{v \in V} \sum_{i=1}^n E_2^V s_i^v \quad (5)$$

$$\pi_5 = \sum_{v \in V} C_v Y_v \quad (6)$$

$$\pi_6 = \sum_{v \in V} L_v / Y_v \quad (7)$$

3.2.3. Model constraints

To capture the essence of the cold chain, constraints are applied to routes to address the business, cold chain and operational needs. These constraints are divided into six constituents: routing, cumulative, time, capacity, labour and variables.

3.2.3.1. Routing constraints. Routing constraints describe the relationship between the binary attributes used to preserve the integrity and meaning of the route. Equation (8) ensures that each node is served by exactly one vehicle and either leads to another node or returns to the depot. As denoted in Eq. (9), each node is preceded by another node or depot. Equation (10) indicates that a vehicle is employed only if it departs from the depot to serve a node. Similarly, Eq. (11) indicates that a vehicle is employed only if it returns to the depot after serving a node. As denoted in Eq. (12), a vehicle is employed only if it serves any node, where M is a sufficiently large number.

$$\sum_{v=1}^V \sum_{j=1}^{n+1} X_{ij}^v = 1 \quad i = 0, \dots, n \quad (8)$$

$$\sum_{v=1}^V \sum_{i=0}^n X_{ij}^v = 1 \quad j = 1, \dots, n \quad (9)$$

$$\sum_{i=1}^n X_{0i}^v = Y_v \quad v \in V \quad (10)$$

$$\sum_{i=1}^n X_{i,n+1}^v = Y_v \quad v \in V \quad (11)$$

$$M \times Y_v \geq \sum_{i=0}^n \sum_{j=1}^{n+1} X_{ij}^v \quad v \in V \quad (12)$$

3.2.3.2. Cumulative constraints. The numbers of attributes in Table 4 represent the cumulative variables that are incurred at each step of a path. The mechanism of this cumulation follows a general pattern: If node i leads to j , the cumulative at j is equal to the cumulative at i plus the value incurred at j as well as any potential slack value. Based on the formulation of Perron and Furnon (2023), the cumulative constraints of the models are described by Eqs. (13)–(18). If vehicle v visits node j after i , the difference in the total travel distance is the travel distance between i and j , as described in Eq. (13). If vehicle v visits node j after i , the difference in the cumulative travel time is the travel time between i and j as well as the waiting time, represented by the slack time s_j^v , as described in Eq. (14). If vehicle v visits node j after i , the difference in the cumulative delivery weight is the weight incurred at node j , as described in Eq. (15). If vehicle v visits node j after i , the difference in the cumulative delivery volume is the volume incurred at node j , as described in Eq. (16).

Equation (17) indicates that if vehicle v visits node j after i , the difference in the cumulative pickup weight is the weight incurred at node j . If vehicle v visits node j after i , the difference in the cumulative pickup volume is the volume incurred at node j , which is described in Eq. (18).

$$X_{ij}^v \left(TD_i^v + d_{ij} - TD_j^v \right) = 0 \quad v \in V; i = 0, \dots, n; j = 1, \dots, n+1 \quad (13)$$

$$X_{ij}^v \left(TT_i^v + t_{ij} + s_j^v - TT_j^v \right) = 0 \quad v \in V; i = 0, \dots, n; j = 1, \dots, n+1 \quad (14)$$

$$X_{ij}^v \left(DW_i^v + w_j^G - DW_j^v \right) = 0 \quad v \in V; i = 0, \dots, n; j = 1, \dots, n \quad (15)$$

$$X_{ij}^v \left(DU_i^v + u_j^G - DU_j^v \right) = 0 \quad v \in V; i = 0, \dots, n; j = 1, \dots, n \quad (16)$$

$$X_{ij}^v \left(PW_i^v + w_j^P - PW_j^v \right) = 0 \quad v \in V; i = 0, \dots, n; j = 1, \dots, n \quad (17)$$

$$X_{ij}^v \left(PU_i^v + u_j^P - PU_j^v \right) = 0 \quad v \in V; i = 0, \dots, n; j = 1, \dots, n \quad (18)$$

3.2.3.3. Time constraints. These constraints describe the impact of business and the cold chain time window as well as the impact of service time on the attributes. Slack time includes waiting or service time; thus, it is not less than the service time, which is proportional to the volume and weight incurred, as shown in Eq. (19). The completion time of each node must be within the customer requested time window, as denoted in Eqs. (20) and (21). The completion time of each node must satisfy the cold chain time window, which is denoted in Eqs. (22) and (23).

$$s_i^v \geq (u_i^G + u_i^P) / V_t^{max} \quad i = 1, \dots, n \quad (19)$$

$$TT_i^v \leq LT_i^C \quad v \in V; i = 0, \dots, n \quad (20)$$

$$TT_i^v \geq ET_i^C \quad v \in V; i = 0, \dots, n \quad (21)$$

$$TT_i^v \leq LT_i^{CC} \quad v \in V; i = 0, \dots, n \quad (22)$$

$$TT_i^v \geq ET_i^{CC} \quad v \in V; i = 0, \dots, n \quad (23)$$

3.2.3.4. Reversible, capacity and loading constraints. This constituent describes the constraints and mechanisms in the capacity of the vehicle during a trip, denoted by Eqs. (24)–(27). Each vehicle leaves the depot on a delivery journey containing the goods it must deliver, the weight of which is $\sum_{k=0}^n w_k^G \sum_{j=1}^{n+1} X_{ij}^v$. The capacity is then updated by subtracting the weight of the delivered goods from the weight of the pickup, as described in Eq. (24). Equation (25) ensures that the total weight of the cargo does not exceed the weight capacity of the vehicle. An analogous description of the volume is given by Eqs. (26) and (27).

$$W_i^v = PW_i^v - DW_i^v + \sum_{k=0}^n w_k^G \sum_{j=1}^{n+1} X_{ij}^v \quad v \in V; i = 0, \dots, n \quad (24)$$

$$w_i^v \leq W_v \quad v \in V; i = 0, \dots, n; j = 1, \dots, n \quad (25)$$

$$V_i^v = PU_i^v - DU_i^v + \sum_{k=0}^n u_k^G \sum_{j=1}^{n+1} X_{ij}^v \quad v \in V; i = 0, \dots, n; j = 1, \dots, n \quad (26)$$

$$V_i^v \leq W_v \quad v \in V; i = 0, \dots, n; j = 1, \dots, n \quad (27)$$

3.2.3.5. Labour constraints. To model the labour required for routing, it is necessary to define the labour demand at each node. As denoted in Eq. (28), the labour requirement cannot exceed the maximum labour capacity of the vehicle. Each worker cannot handle more volume than is necessary, and there must be a minimum of one labourer (the driver), as shown in Eq. (29). Likewise, each labourer cannot handle more weight than their capacity, as denoted in Eq. (30).

$$L_v \leq L_v^{max} \quad v \in V \quad (28)$$

$$L_v \geq \lceil DU_{n+1}^v / U_l^{max} \rceil \quad v \in V \quad (29)$$

$$L_v \geq \lceil PU_{n+1}^v / U_l^{max} \rceil \quad v \in V \quad (30)$$

3.2.3.6. Variable constraints. Finally, this section describes the standard initial conditions of the variables. The binary nature of X_{ij}^v and Y^v is expressed by Eqs. (31) and (32). Every cumulative variable is a physical quantity and therefore non-negative, as are denoted by Eqs. (33)–(35). Apart from slack time, which can be non-zero at the start, every other cumulative variable starts at zero, as denoted in Eq. (36).

$$X_{ij}^v \in \{0, 1\} \quad v \in V; i = 0, \dots, n; j = 1, \dots, n+1 \quad (31)$$

$$Y_v \in \{0, 1\} \quad v \in V \quad (32)$$

$$DU_i^v, PU_i^v, DW_i^v, PW_i^v \geq 0 \quad v \in V; i = 1, \dots, n \quad (33)$$

$$S_i^v, W_i^v \geq 0 \quad v \in V; i = 0, \dots, n \quad (34)$$

$$TD_i^v, TT_i^v \geq 0 \quad v \in V; i = 1, \dots, n+1 \quad (35)$$

$$DU_0^v, PU_0^v, DW_0^v, PW_0^v, TD_0^v, TT_0^v = 0 \quad v \in V \quad (36)$$

4. Simulation case study

To validate the performance of the RCCVRP model, it was applied to a company through a simulation. The company under study was a Hong Kong-based conglomerate LSP that specialises in both air- and sea-based international freight transportation services. Their areas of focus include aviation, automotive, chemical, fashion, healthcare, project cargo, retail and technology transportation. The demand for CCL has grown rapidly in recent years and was most pronounced during the COVID-19 pandemic, resulting in an unprecedented demand for pharmaceuticals. In response to this, the studied company established a 'Pharma Cold Logistics' brand in 2018 to focus on providing flexible and door-to-door CCL services, specifically in cold chain packaging and handling solutions. They have expertise in managing temperature-sensitive freight shipments and offer the 'At Door Pick and Pack Service', which initiates cold chain management at the shipper's location. Their temperature-controlled warehouses and refrigerator trucks are shown in [Figure 3](#).

For energy savings and logistical reasons, it is common for practitioners in the CCL industry to utilise passive rather than active packaging as a cold chain solution. Moreover, disposable passive packaging is often chosen over reusable packaging. Therefore, sustainability is difficult to achieve in terms of waste and resource use. Recognising the importance of sustainability, the company has prioritised reducing its environmental footprint within its CCL operations and has taken initiatives to improve not only economic viability but also environmental sustainability and social responsibility. By applying the proposed framework and the concept of sustainability, the company aims to create a strong foothold in cold chain markets and gain potential sustainability-conscious customers. To address sustainability and the resulting challenges in reverse logistics, the RCCVRP model was implemented using simulated data from the case company, and the details highlighting the key stages of the proposed model are shown in [Figure 4](#).

4.1. Stage 1: data collection and consolidation

Data from six areas were gathered in this case study: (i) labour, (ii) transportation, (iii) fuel, (iv) emissions, (v) cold chain packaging and (vi) transit. The case study utilised data



Figure 3. A temperature-controlled warehouse and refrigerator truck of the case company.

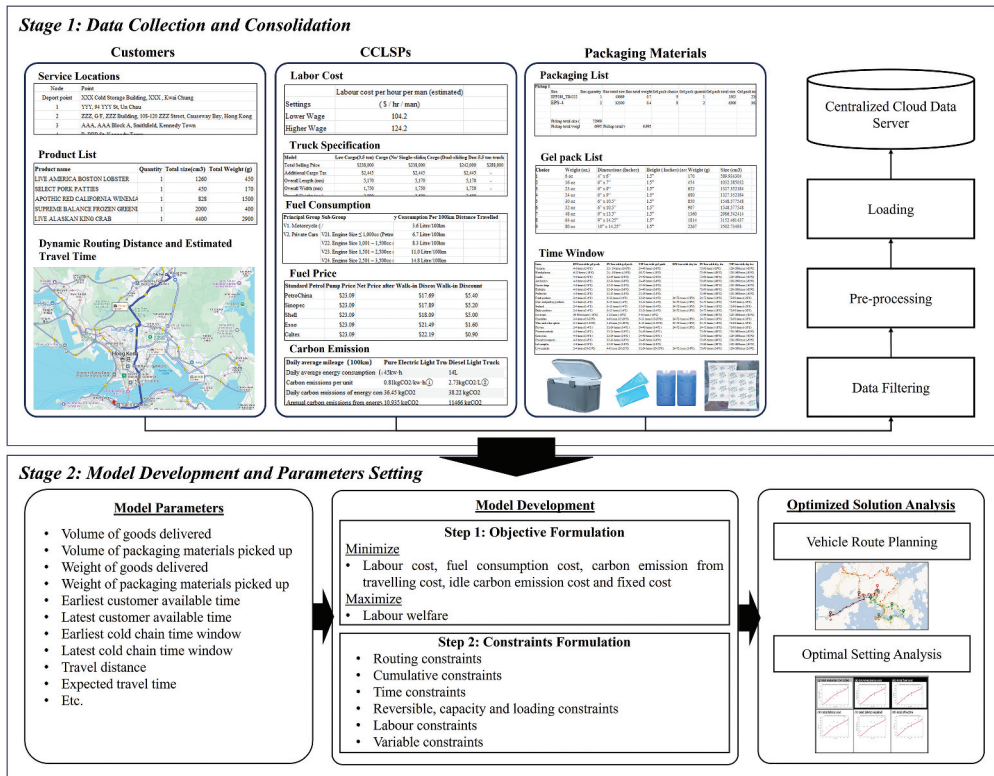


Figure 4. RCCVRP model of the case study.

obtained from 2022 to 2023 via the Internet. The labour information was considered the labour cost per unit time, which consisted of the wages of the part-time workers involved in the routing process at an hourly rate. The transportation information consisted of the specifications of the refrigerated trucks employed and included the cost, capacity and fuel consumption. Fuel information comprised fuel price. The emission information consisted of the CO₂ emissions through fuel consumption, and the resultant emission cost was taken as the carbon tax, which consisted of the EU carbon tax policy and the corresponding exchange rate to be converted back to HKD. This was taken to be 90€ per metric ton. The cold chain packaging information included the product, EPS box, packaging, gel pack and cold chain time window. Examples of an EPS box and a gel pack are shown in Figure 5. The transit information was derived from a list of service locations and drafted to represent the depot and delivery locations. The resulting transit data were collected in the form of distance and time matrices using Google Maps via the Google Script API. The collected data are summarised in Table 2.

4.2. Model development

The collected raw data were then converted into variables to be fed into Google OR-Tools, which were used to model and provide a heuristic solution for the RCCVRP model. The objective was calculated based on labour cost, travel distance, travel time, slack time,



Figure 5. Example of the EPS box and gel pack.

Table 2. Data collection.

Category	Details	Description
Labour	Labour	Wage of part-time workers (\$/hour)
Transportation	Truck specification	Cost and dimensions of the truck
Fuel	Fuel consumption	Fuel consumption of the truck (L/100km)
	Fuel price	Fuel price (\$/L)
Emission	Carbon emission	Carbon emission per unit fuel consumed (kgCO ₂ /L)
	Carbon tax	Carbon tax based on CO ₂ emission based on EU standard
	Exchange rate	Exchange rate from Euro to HKD
Cold chain packaging	Passive packaging	Specifications of the passive packaging
	Time window	The cold chain time windows of the products
Transit	Service location	Location of the service nodes
	Distance matrix	Distance between the service locations
	Time matrix	Travel duration between the service locations

number of vehicles used and social sustainability in terms of labour welfare. The distance and time matrices were obtained from the Google Maps API and had units of 100 m and min, respectively. The fixed cost consisted of wear, maintenance and emission component costs and was in units of HKD (\$). The conversion units for labour cost consisted of labour cost, which was the wage at an hourly rate. The conversion units for fuel cost consisted of fuel price, which was measured in cost per litre, and fuel consumption, which was measured in litres of fuel consumed per 100 km travelled. The conversion units for travelling emissions consisted of fuel consumption and carbon emissions, which were measured in kilograms of CO₂ per litre of fuel consumed and carbon tax in euros per metric ton, respectively, and the exchange rate from euro to HKD. The conversion units for idle emissions consisted of carbon emissions measured in kilograms of CO₂ per hour, carbon tax and exchange rate. The fixed cost included the sum of truck wear, maintenance and starting emission costs. Truck wear cost was calculated as the price of the truck divided by its expected lifespan. The maintenance cost consisted of the maintenance cost and maintenance period, which was taken to be once per month. The starting emission cost was computed as the product of carbon emission cost and the carbon tax. The unit rates of each variable are listed in Table 3.

The conversion scale λ_p was then chosen to ensure all factors were converted into the same unit. The calculations of the unit cost for each factor are summarised in Table 4.

Table 3. Unit rates of each variable.

Variable	Light Truck	Medium Truck	Input Variables of Factor p (IV_p)
Labour cost	\$104.2/hour	\$124.2/hour	IV_1
Fuel price		\$20.8/litre	IV_2
Fuel Consumption	11.1 litre/100km	20.1 litre/100km	$IV_2 \cap V_3$
Traveling emission		2.73 kg/litre	$IV_3 \cap V_4 \cap V_5$
Idle emission		1.5 kg/litre	
Starting emission		0.8458 kg/litre	
Carbon Tax		90€/metric tonne	$IV_3 \cap V_4 \cap V_5$
Exchange rate		1€ = \$8.486	$IV_3 \cap V_4 \cap V_5$
Truck cost	\$238,000	\$288,000	IV_5
Truck Life Span		5,475 days	IV_5
Maintenance cost		900	IV_5
Maintenance period		30 days	IV_5

4.3. Parameters setting

Some parameters are prone to change to reflect the dynamic nature of the business of the company. Specifically, this study considered four parameter settings based on truck type and the cold chain time window. Table 5 outlines the parameter configurations related to trucks, prices and cold chain time windows. These settings were selected to evaluate the effect of changes in truck capacities, pricing strategies and cold chain time windows on the objectives. Table 6 summarises the detailed optimisation parameters corresponding to each setting listed in Table 5. For example, setting 1 in Table 5 introduces a 3.3-ton truck with a lower pricing strategy and a 3-h cold chain delivery time window. Table 6 presents the configuration, which includes a vehicle weight capacity of 1,090 kg, a total volume of 7,587,792 cm³ and cost, environmental, and delivery parameters retrieved from the API and demand volume list.

5. Results and discussion

This section presents the detailed results of the case study and is divided into three parts: (i) a summary of results, which provides the key performance indicators (KPIs) that facilitate comparison between settings, as well as strategy recommendations to improve the operation of the case company; (ii) detailed results of the best setting; and (iii) summary and implications.

5.1. Experimental results

This section presents the experimental results for each objective. Table 7 presents the comparison results with arithmetic means and standard deviations. By setting the equal weightings on the importance of each factor, that is, $\delta_p = 0.167$, we found that setting 3 (truck type = 3.3 ton, TW = 2) provided minimum values for the costs of labour, fuel, travel emission, and idle emission as well as the fixed cost. For social sustainability in terms of labour welfare, a higher labour ratio in the results is preferred so that more labourers can work simultaneously to share the workload. Setting 4 (truck type = 5.5 ton, TW = 2) provided the maximum value in terms of the labour ratio, at 1.46. For the overall objective, the results indicated that setting 3 (truck type = 3.3 ton, TW = 2) was the best setting. The mean of the

Table 4. Calculation of unit cost for each cost factor.

	Rate (\$/hour)	=	Cost per unit time (\$/hour)	Conversion λ_1	Unit Cost (\$/min)
Labour cost					
– Light Truck	104.2		104.2	60	1.7367
– Medium Truck	124.2		124.2	60	2.07
Fuel consumption cost					
	Fuel Price (\$/L)	x	Fuel consumption (L/100km)	Conversion λ_2	Unit Cost (\$/100m)
– Light Truck	20.8		11.1	1,000	0.2309
– Medium Truck	20.8		20.1	1000	0.4181
Carbon emission from travelling cost					
	CO ₂ emission (kg/L)	x	Fuel consumption (L/100km)	Conversion λ_3	Unit Cost (\$/100m)
– Light Truck	2.73		11.1	1,000	0.0231
– Medium Truck	2.73		20.1	1,000	0.0419
Idle carbon emission					
	CO ₂ emission (kg/hour)	x	Carbon tax (€/1000kg)	Conversion λ_4	Unit Cost (\$/min)
1.5			90	60	0.0191
Fixed cost					
	Truck wear (\$)	+	Maintenance (\$)	Conversion λ_5	Unit Cost (\$)
– Light Truck	238,000/5,475 = 43,470		900/30 = 30	1	74.116
– Medium Truck	288,000/5,475 = 52,602		900/30 = 30	1	83.248

Table 5. Parameters setting for comparison.

Setting	Truck Type	Price	Cold Chain Time Window
1	3.3-ton truck	Low	3 hours
2	5.5-ton truck	High	3 hours
3	3.3-ton truck	Low	2 hours
4	5.5-ton truck	High	2 hours

Table 6. OR-Tools parameters.

OR-Tools parameters	Value			
	Setting 1	Setting 2	Setting 3	Setting 4
Vehicle weight capacity (kg)	1,090	1,300	1,090	1,300
Vehicle volume capacity (cm ³)	7,587,792	14,000,000	7,587,792	14,000,000
Vehicle fixed cost (\$)	74.1163	83.2487	74.1163	83.2487
Labour cost (\$/min)	1.7367	2.0700	1.7367	2.0700
Fuel cost (\$/100m)	0.2309	0.4181	0.2309	0.4181
Transit emission cost (\$/100m)	0.231	0.419	0.231	0.419
Cold chain time windows (min)	{0,180}	{0,180}	{0,120}	{0,120}
Number of nodes			6–13	
Number of vehicles			5–10	
Idle emission cost (\$/min)			0.0191	
Maximum labour per vehicle			3	
Maximum volume served per minute (cm ³)			250,000	
Volume handled per labour (cm ³)			500,000	
Travel distance (100m)			Retrieved from Google Script API	
Travel time (min)			Retrieved from Google Script API	
Delivery volume (cm ³)			Retrieved from the delivery demand volume list	
Delivery weight (kg)			Retrieved from the delivery demand weight list	
Pickup volume (cm ³)			Retrieved from the pickup demand volume list	
Pickup weight (kg)			Retrieved from the pickup demand weight list	
Customer time windows (min)			Retrieved from the customer time window list	

Table 7. Comparison of the four parameter settings.

Factor	Truck Type	μ mean		σ s.d.	
		3.3 ton	5.5 ton	3.3 ton	5.5 ton
Labour Cost	TW = 3	871.82	1076.10	235.55	304.72
	TW = 2	720.11	884.63	189.45	242.05
Fuel Cost	TW = 3	488.45	847.73	100.37	177.46
	TW = 2	470.29	832.11	112.56	183.59
Travel Emission Cost	TW = 3	48.87	84.96	10.04	17.78
	TW = 2	47.05	83.39	11.26	18.40
Idle emission Cost	TW = 3	0.68	0.73	0.44	0.43
	TW = 2	0.29	0.29	0.08	0.08
Fixed Cost	TW = 3	259.41	279.48	69.71	69.71
	TW = 2	248.82	273.53	74.72	76.08
Labour Ratio	TW = 3	1.36	1.43	0.26	0.32
	TW = 2	1.43	1.46	0.25	0.23
Overall Objective	TW = 3	835.30	1145.21	187.09	251.66
	TW = 2	743.99	1037.71	180.09	249.41

overall objective in setting 3 was 743.99, which was the lowest among the four settings. This setting also provided the best performance for all the cost factors. The detailed results of the best setting (setting 3 with truck type = 3.3 ton, TW = 2) are described below.

5.2. Optimal setting

This section describes the details of the optimal setting, that is, setting 3 with a truck type of 3.3 ton and a time window of 2 h. According to the results, setting 3 outperformed the others in terms of the overall objectives by 12.7%, 53.93% and 39.48%. Although the labour ratio for setting 3 was 5% lower than that for setting 1, the other factors of improvement were significantly higher than those of setting 1. Table 8 compares the parameter changes using optimal setting 3 and summarises the differences between the optimal and alternative settings.

An example of the generated route is shown in Figure 6, and the results are summarised in Figure 7. First, every KPI showed a positive and approximately linear relationship between the number of orders and the value. This indicates that it is reasonable to consider the marginal difference per unit order. Second, there was a significant variation in the values. For example, in this study, the total travel distance with 11 nodes was greater than that with 13 nodes. Similar results were also observed for other KPIs, except the total labour required, idle emissions and slack time. This was expected because of the

Table 8. Comparative analysis of parameters using optimal setting 3.

Factor	μ mean Setting 3	Relative Differences Compared to Setting 3		
		Setting 1	Setting 2	Setting 4
Labour Cost	720.11	21.07%	49.44%	22.85%
Fuel Cost	470.29	3.86%	80.26%	76.94%
Travel Emission Cost	47.05	3.87%	80.57%	77.24%
Idle Emission Cost	0.29	134.48%	151.72%	0.00%
Fixed Cost	248.82	4.26%	12.32%	9.93%
Labour Ratio	1.43	– 4.90%	0.00%	2.10%
Overall Objective	743.99	12.27%	53.93%	39.48%

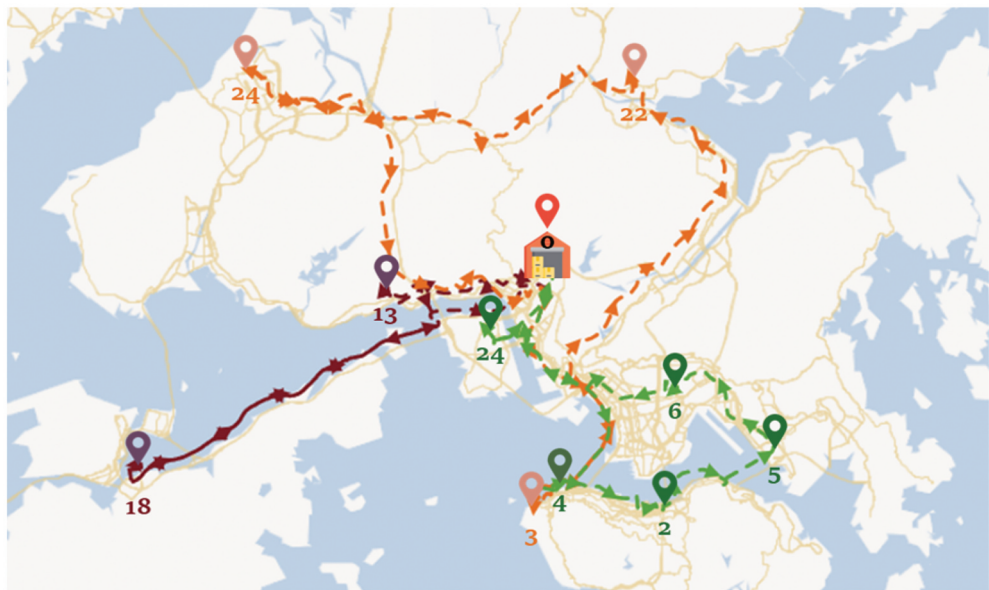


Figure 6. Vehicle route 5 of setting 3 illustrated in Google map.

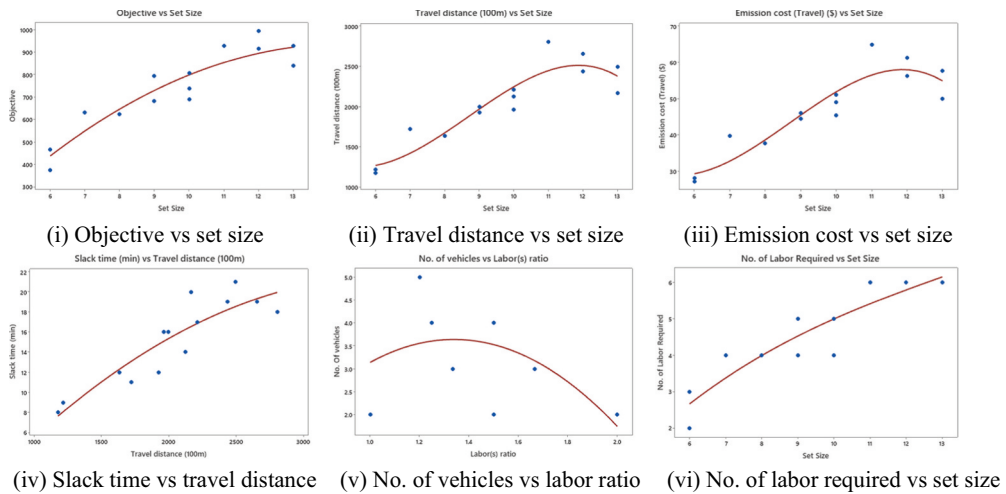


Figure 7. Selected KPIs of the best setting.

complexity of vehicle routing. Third, the total labour required a step function increase, which was expected as the workers employed are discrete and depend only on the load capacity.

5.3. Summary and implications

Based on the comparative study of the setting parameters, a tighter time window significantly enhanced performance, whereas the size of the trucks showed no notable impact. This unexpected finding suggests that neither the cold chain time window nor truck size is a limiting factor across nearly all the metrics assessed. Implementing stricter conditions, such as shorter time windows, can be advantageous for both the company and its customers. Overall, the case study effectively demonstrated the capability of the RCCVRP model to optimise CCL operations while adhering to environmental sustainability constraints, particularly in (i) gas emission reduction and elimination and (ii) multi-objective routing that balances customer, business and environmental needs. The academic contributions of this study are as follows. First, the RCCVRP model facilitates reverse logistics operations by recycling reusable passive packaging between CCL providers and customers. This success serves as a proof of concept for reverse logistics, aligning with the goals of a circular economy and waste minimisation. Second, the model addresses a critical gap in the literature on multi-objective optimisation within CCL and sustainability, expanding the existing body of knowledge. From a practical perspective, our model effectively eliminates packaging waste and gas emissions in cold chain operations, significantly contributing to the sustainability goals of the company in the case study. Furthermore, the successful implementation of the RCCVRP model in the case study provides a practical methodology for the industry to adopt reverse logistics in a cold chain context while maintaining high levels of customer satisfaction and addressing business and environmental concerns.

Moreover, fair and sustainable working conditions reinforce a company's commitment to social responsibility, and the RCCVRP model considers the maximum labour capacity and handling volume of workers. Finally, this model offers a systematic approach for determining and analysing parameters during model construction, equipping companies with the necessary tools to evaluate the effectiveness of their operational settings. Based on this study, the proposed RCCVRP model can improve CCL by optimising environmental, business and social responsibility objectives in practice.

6. Conclusions

This study significantly contributes to the research domain of sustainable CCL and lays the groundwork for the development of the proposed RCCVRP model. Specifically, we introduced a multi-objective routing model designed to effectively address the complexities of simultaneous reverse logistics in cold chains, while balancing business, operational and sustainability considerations. Furthermore, we developed a novel methodology for determining the weight factors for each objective, addressing the existing gap in data mining that limits its practical applications in the field. To validate our approach, we conducted a comprehensive simulation case study, demonstrating that the proposed methodology serves as an effective scheduling system for CCL at both the strategic and tactical levels. This addressed the practical needs fulfilled by an enhanced reverse logistics framework that not only minimises waste and carbon emissions but also aligns with broader sustainability goals. However, this model has two limitations. First, it lacks sensitivity analysis of the significance of the input parameters, which may result in inaccuracies in the model and affect its performance. Second, the study assumed preloaded goods and overlooked the operational times at the depot, which may not accurately reflect real-world scenarios. To address these limitations, two directions for future research are suggested. (i) Conducting a sensitivity analysis of the impact of the significance of the parameters. This analysis could highlight the necessity for precise measurements, or conversely, indicate that less stringent figures might suffice, thereby reducing computational complexity and alleviating pressure on the company. (ii) Relaxing the preloaded assumption and incorporating the dynamic quality of goods into the optimisation model to yield better insights into the effects of routing and cooling materials on product quality and customer satisfaction in the cold chain.

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